D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

# CRISPR–Cas9

**August 2022** | DM 0822 | Q8 (Short note (FOCUS 2022–2026 · NCCN 2026)) | **7 marks** | Target ~13 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

## 1. Introduction & exam framing

This answer addresses: **CRISPR–Cas9**

For a **7-mark** university answer, structure the response as: definition/epidemiology → biology/hallmarks → diagnosis & biomarkers → staging/risk → management algorithm with doses → key clinical trials with endpoints → toxicities/supportive care → **NCCN 2026 updates** → Indian/NCG context → summary table. Draw at least one flowchart and one mechanism diagram.

Marking examiners reward: accurate mechanisms, named trials with phase and primary endpoint, practical doses, Category evidence (NCCN/ESMO), and a clear final recommended pathway.

- Start with a one-line definition + why the topic is ‘recent advances’ relevant.
- Always state predictive vs prognostic biomarkers.
- Quote Category 1 preferred regimens from NCCN 2026/2027 when applicable.
- End with 5 high-yield exam pearls.

## 2. Pathobiology, hallmarks & mechanism

Cancer biology for this topic should be linked to Hanahan–Weinberg hallmarks (sustained proliferation, evading growth suppressors, resisting cell death, enabling replicative immortality, induction of angiogenesis, activating invasion/metastasis, avoiding immune destruction, deregulating cellular energetics, genome instability, tumor-promoting inflammation).

Where a driver oncogene/fusion/epigenetic lesion is known, explain the pathway from ligand/receptor → intracellular kinase cascade → transcription → phenotype, and mark the drug target with an arrow in your diagram.

| | |
|---|---|
| **Driver / target** | State oncogene, fusion, receptor, or immune checkpoint involved |
| **Pathway** | MAPK / PI3K-AKT-mTOR / JAK-STAT / DDR / BCR signaling / angiogenesis |
| **Hallmark effect** | Proliferation, survival, angiogenesis, immune escape, or DNA repair addiction |
| **Drug class** | mAb / TKI / ADC / BiTE / CAR-T / CPI / PARPi / IMiD / BH3 mimetic |
| **Resistance** | Secondary mutation, bypass track, phenotypic switch, antigen loss, TME suppression |

*Figure: mechanism / pathway map*

**EXAM PEARLS / HALLMARKS**

- ASCO/CAP: IHC 0/1+/2+/3+; ISH if 2+; HER2-low = IHC 1+ or 2+/ISH–
- Draw the pathway once; label drug targets — examiners love this.
- Always mention one primary resistance mechanism.

## 3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.
- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).
- Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.

**Work-up algorithm (draw this in the answer booklet)**

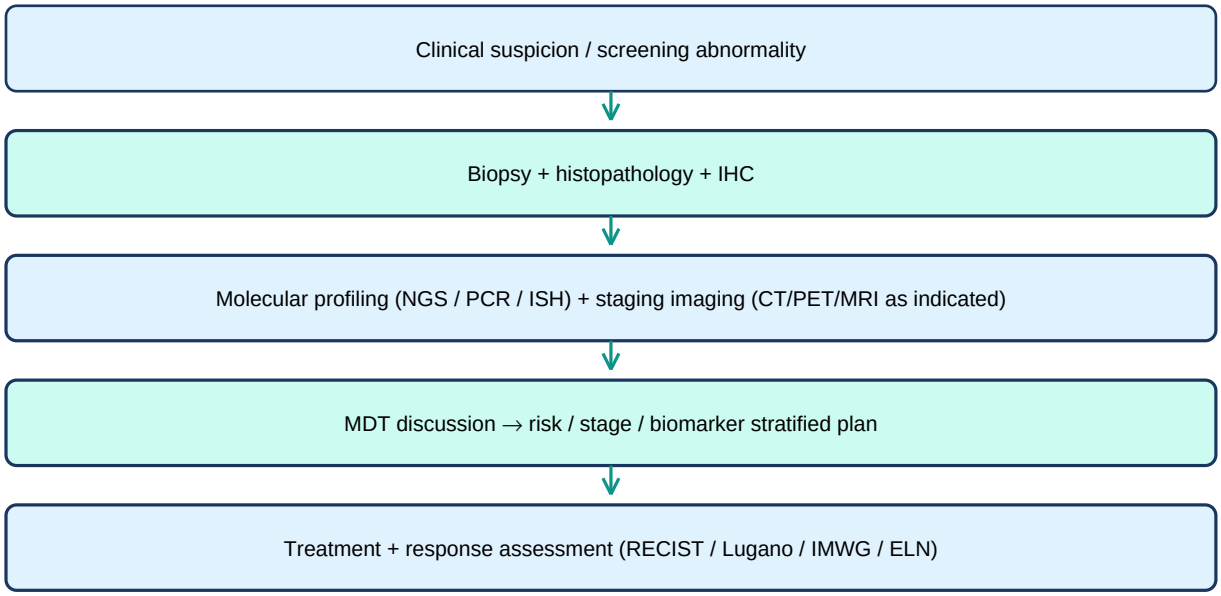


*Figure: stepwise exam algorithm*

## 4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

- Localized disease: multimodality — surgery/RT/systemic as per disease chapter.
- Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).
- Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.
- Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).
- Re-biopsy / liquid biopsy at progression to map resistance.

**Management flowchart (reproduce in exam)**

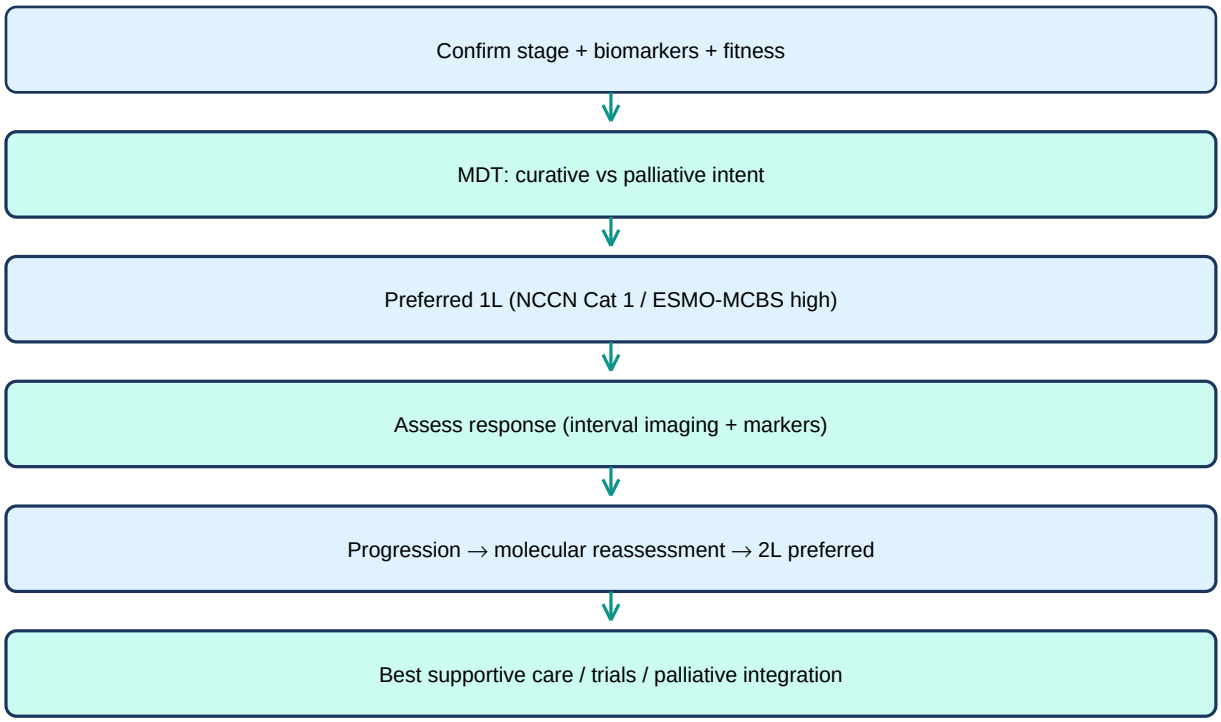


*Figure: stepwise exam algorithm*

| Drug / regimen | Dose / schedule (exam pearls) |
|---|---|
| Pembrolizumab | 200 mg IV Q3W or 400 mg Q6W |
| Nivolumab | 240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen |
| Olaparib | 300 mg PO BID (tablets); adjust for toxicities/renal; OlympiA adjuvant 1 year |
| Trastuzumab | 8 mg/kg → 6 mg/kg IV Q3W (or SC fixed dose) |

*Table: doses and regimens*

## 5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

- Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.
- Distinguish statistical vs clinical significance; note crossover and subgroup caveats.
- For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

| Trial | Key result / exam takeaway |
|---|---|
| HERA / BCIRG-006 / NSABP B-31 / N9831 | Adjuvant trastuzumab OS/DFS benefit |
| CLEOPATRA | Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC |
| FLAURA | Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L |
| FLAURA2 | Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred |
| MOSAIC / C-07 / IDEA | Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX |
| ATOMIC | FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred |
| ZUMA-1 / JULIET / TRANSCEND | Axi-cel; tisa-cel; liso-cel LBCL |
| ZUMA-7 / TRANSFORM / BELINDA | 2L CAR-T vs SOC |

*Table: landmark clinical trials*

## 6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

- Anticipate class toxicities and counsel patients; use CTCAE grading.
- Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.
- IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.
- ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.
- TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.
- Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.

**Toxicity management loop**

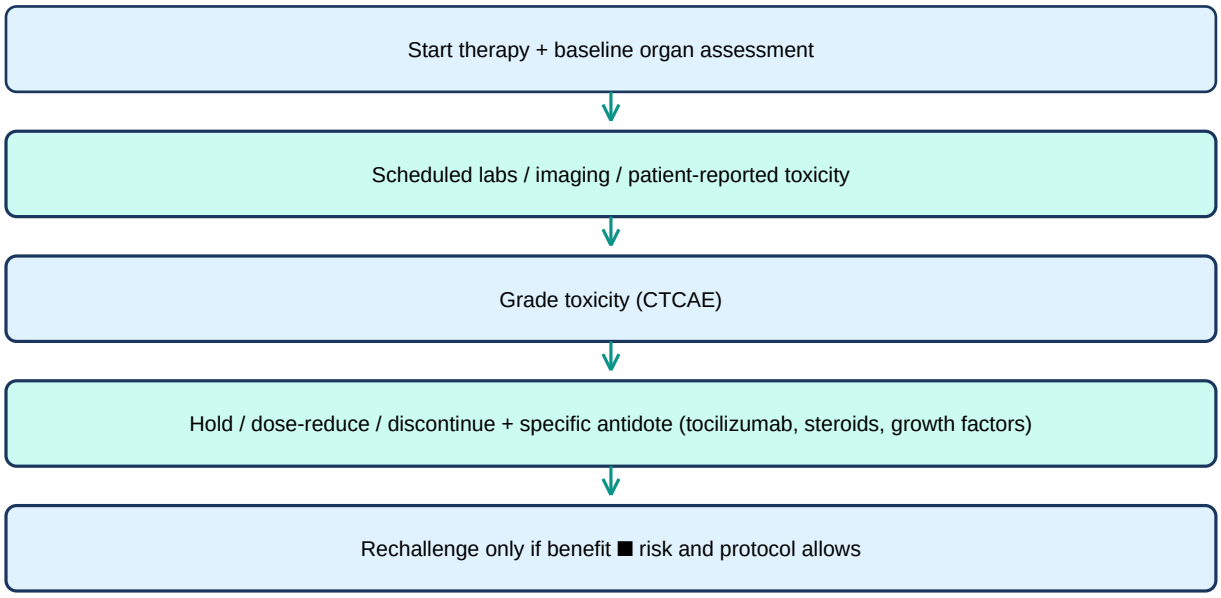


*Figure: stepwise exam algorithm*

## 7. NCCN 2026 / 2027 updates (high-yield for ‘recent advances’)

Paper IV rewards currency. Explicitly write: **“As per NCCN Guidelines updates in 2026/2027…”** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

- State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.
- If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

**NCCN 2026 / 2027 UPDATES (include in answer)**

- *Referenced versions:* NCCN Guidelines® (latest 2026/2027 disease-specific version)
- Cite the latest disease-specific NCCN Guidelines® (2026/2027 versions) and NCCN Drugs & Biologics Compendium for category ratings.
- Mention Category 1 vs 2A/2B and preferred vs other recommended vs useful in certain circumstances.
- Cross-check FDA label, institutional availability (India), and National Cancer Grid pathways where relevant.

**EXAM PEARLS / HALLMARKS**

- NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).
- Category 1 = high-level evidence + uniform NCCN consensus.

## 8. Special situations, differentials & practical issues

Discuss differentials, mimics, and when to refer to radiation oncology, surgery, palliative care, genetics, or cell-therapy centers.

Include brief notes on health economics (cost-effectiveness, biosimilars) and clinical trial enrollment as standard options.

- MDT documentation improves outcomes and exam completeness.
- Survivorship care plans: late effects, second malignancies, cardiac, endocrine, psychosocial.
- Real-world evidence and digital tools increasingly complement RCT data (relevant to 2026 Paper IV themes).

## 9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

- Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.
- Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.
- Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.
- Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

**EXAM PEARLS / HALLMARKS**

- Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)
- Mention number-needed-to-treat conceptually if space permits.

## 10. Summary checklist (write this last page in the booklet)

Close the 7-mark answer with a crisp numbered summary the examiner can tick.

- Definition + epidemiology one-liner
- Key biology / hallmark diagram
- Essential diagnostics & biomarkers
- Risk/stage stratification
- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

**30-second revision flowchart**

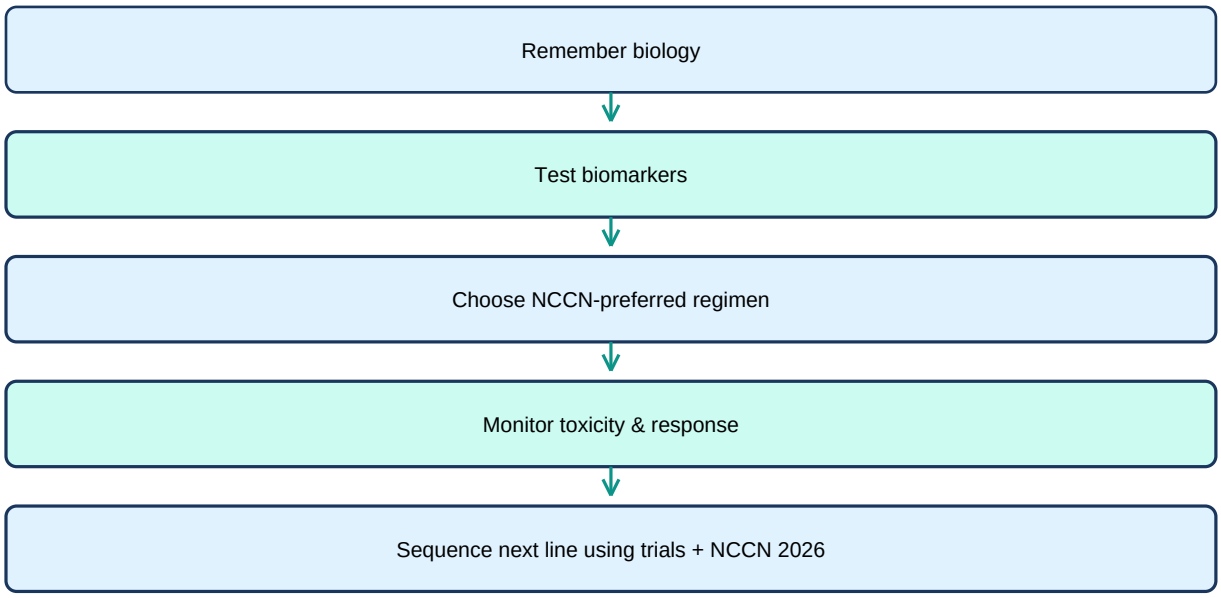


*Figure: stepwise exam algorithm*

**EXAM PEARLS / HALLMARKS**

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say 'NCCN 2026' explicitly for Paper IV.

## 11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.
- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

| Drug / regimen | Dose / schedule (exam pearls) |
|---|---|
| Pembrolizumab | 200 mg IV Q3W or 400 mg Q6W |
| Nivolumab | 240 mg Q2W or 480 mg Q4W; with ipi 1 mg/kg (RCC) or 3 mg/kg (melanoma) per regimen |
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*Table: doses and regimens*

## 12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.
- PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.

- ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

**Clinical algorithm**

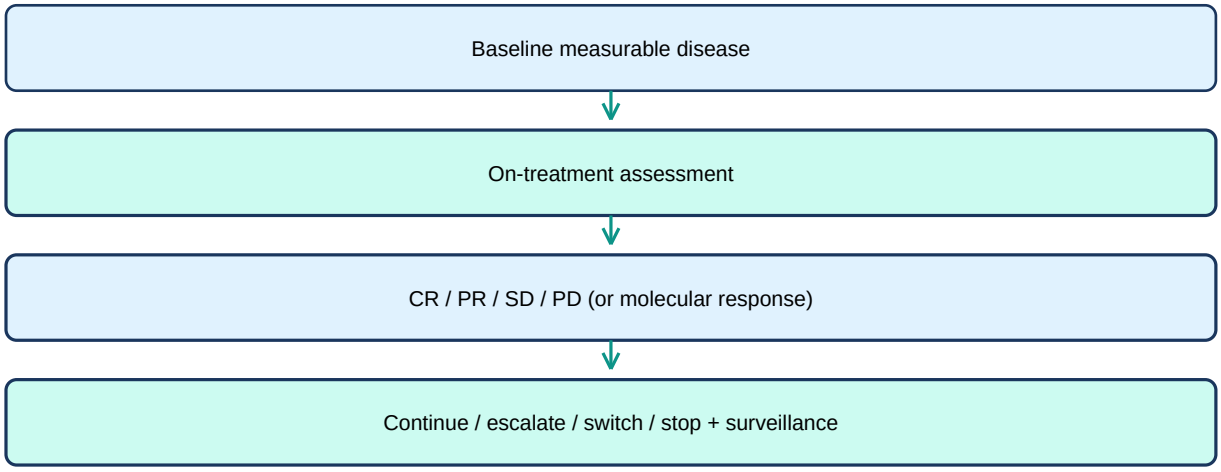


*Figure: stepwise exam algorithm*

## 13. Comparison tables useful for exams

Draw 2×N tables comparing old standard vs new standard; or drug A vs B (efficacy, toxicity, CNS activity, cost/access).

- Column ideas: target, dose, PFS, OS, grade ≥3 AEs, CNS activity, NCCN category.
- Another table: resistance mutation → preferred next drug.

| Trial | Key result / exam takeaway |
|---|---|
| HERA / BCIRG-006 / NSABP B-31 / N9831 | Adjuvant trastuzumab OS/DFS benefit |
| CLEOPATRA | Docetaxel + trastuzumab + pertuzumab → OS ~57 mo 1L HER2+ mBC |
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| MOSAIC / C-07 / IDEA | Adjuvant oxaliplatin; 3 vs 6 mo CAPOX/FOLFOX |
| ATOMIC | FOLFOX/CAPOX + atezolizumab stage III dMMR — NCCN 2026 preferred |

*Table: landmark clinical trials*

## 14. Indian context, NCG & practical exam tips

National Cancer Grid (NCG) resource-stratified guidelines are frequently expected in Indian DM exams. Discuss biosimilars (trastuzumab, rituximab), generic TKIs, and center-based cell therapy availability.

- Document MDT decisions; clinical trial referral when standard options exhausted.
- Counsel financial toxicity transparently; prioritize highest-value Category 1 options.
- Infection prophylaxis algorithms crucial in hematologic topics (antifungal, antiviral, PJP).

## A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

## B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

**Therapeutic eras timeline**

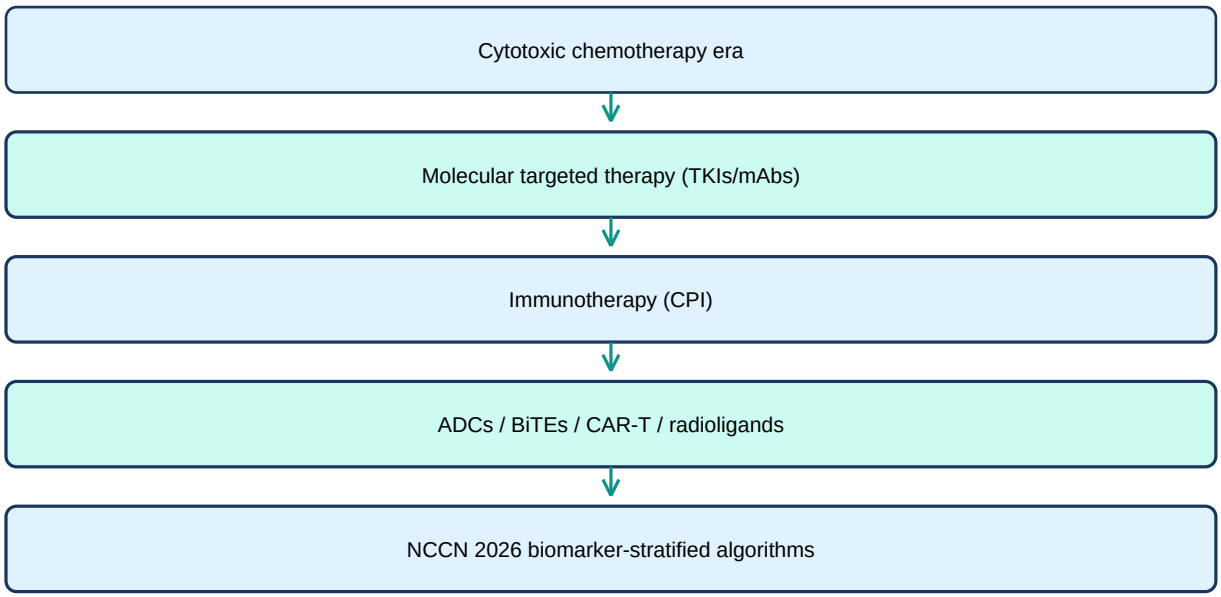


*Figure: stepwise exam algorithm*

## C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

| | |
|---|---|
| **Morphology** | Light microscopy pattern |
| **IHC panel** | Lineage + predictive markers |
| **Cytogenetics** | Karyotype / FISH fusions |
| **NGS** | SNVs, indels, CNAs, MSI, TMB |
| **Integrated Dx** | WHO/ICC + AJCC stage |

*Figure: mechanism / pathway map*

## D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

**Clinical algorithm**

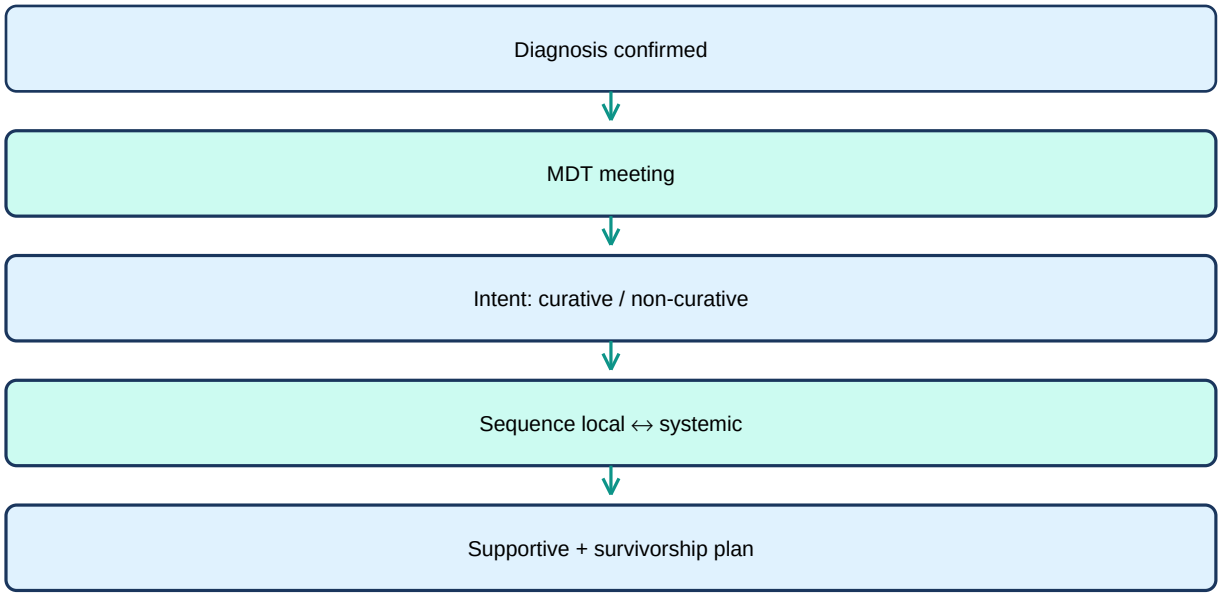


*Figure: stepwise exam algorithm*

## G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

## H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.
- Underline drug doses.
- Box the NCCN 2026 bullets.

**EXAM PEARLS / HALLMARKS**

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.
- NCCN 2026 phrase is expected in Paper IV recent advances.

## K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).
- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

**Clinical algorithm**

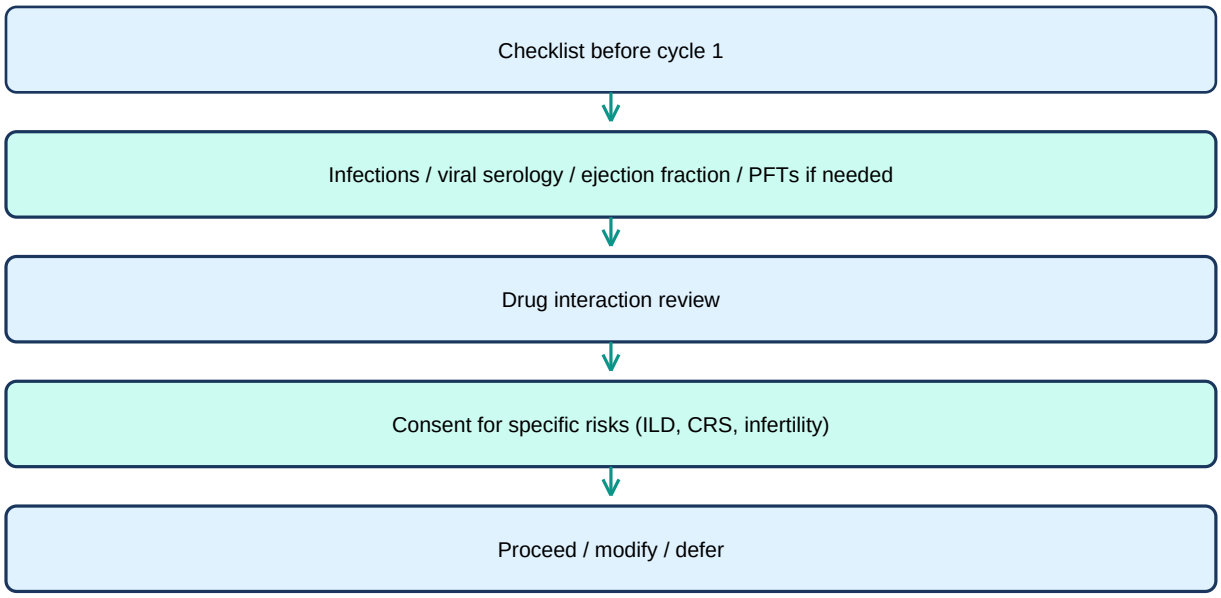


*Figure: stepwise exam algorithm*

## L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.
- Vignette 3: progression with resistance mutation — sequenced next line.
- Vignette 4: severe toxicity — hold, treat, rechallenge decision.

## M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

| | |
|---|---|
| **Quadrant 1** | Hallmark gene / pathway / figure |
| **Quadrant 2** | Must-order tests + staging |
| **Quadrant 3** | 1L / 2L / 3L with doses |
| **Quadrant 4** | Trials + NCCN 2026 + toxicities |
| **Center** | MDT + patient goals |

*Figure: mechanism / pathway map*

**EXAM PEARLS / HALLMARKS**

- Wall chart + flowchart + trial table = visual completeness.
- Write ‘NCCN 2026’ in quadrant 4 explicitly.

## I. Extended notes — resistance, sequencing & future directions

Resistance biology should occupy a dedicated half-page: on-target secondary mutations, bypass signaling, lineage plasticity (e.g., EGFR NSCLC → SCLC), antigen loss after cellular therapy, and immunosuppressive TME changes after CPI.

Sequencing principles: use the agent with largest OS benefit early when toxicities allow; reserve cross-resistant drugs; consider cumulative neuropathy/cardiotoxicity; plan CNS-active agents if brain metastases are present or likely.

Future directions appropriate for 2026 answers: tumor-agnostic indications (NTRK, RET, BRAF, MSI-H, TMB-H, HER2-mut), radioligand therapy, CAR-NK/off-the-shelf products, ctDNA-guided escalation/de-escalation, AI decision-support, and microbiome modulation adjuncts.

Health services angle: real-world evidence validating trial results in older/comorbid patients; digital remote toxicity monitoring; and equity of molecular testing access — themes also reflected in recent Paper IV questions.

- Write one ‘if progression then…’ resistance table.
- Mention at least one tumor-agnostic approval if biology overlaps.
- Close with trial enrollment as a standard option.

## J. Quick-revision flash cards (last 1–2 pages)

Convert the topic into 8–10 flash cards mentally: (1) definition (2) hallmark gene (3) key test (4) 1L preferred (5) dose (6) trial (7) top toxicity (8) NCCN 2026 update (9) Indian/NCG pearl (10) one differential.

- Flash 1: One-line definition
- Flash 2: Driver / pathway
- Flash 3: Mandatory biomarker
- Flash 4: NCCN preferred 1L
- Flash 5: Exact dose
- Flash 6: Landmark phase III + endpoint
- Flash 7: Grade ≥3 toxicity to warn
- Flash 8: 2026 guideline change
- Flash 9: Resistance next step
- Flash 10: Supportive care must-do

**EXAM PEARLS / HALLMARKS**

- If time-poor in the hall: draw algorithm + trial table + NCCN box — captures most marks.
- Keep handwriting large for doses and HRs.

### Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

**NCCN 2026 / 2027 UPDATES (include in answer)**

• *Referenced versions:* NCCN Guidelines® (latest 2026/2027 disease-specific version)

• Cite the latest disease-specific NCCN Guidelines® (2026/2027 versions) and NCCN Drugs & Biologics Compendium for category ratings.

• Mention Category 1 vs 2A/2B and preferred vs other recommended vs useful in certain circumstances.

• Cross-check FDA label, institutional availability (India), and National Cancer Grid pathways where relevant.

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**Disclaimer:** Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.